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Survivability to orbital debris of tape tethers for end-of-life spacecraft de-orbiting

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ABSTRACT

Electrodynamic tethers (EDT) represent one of the possible means to de-orbit defunct satellites from Low-Earth-Orbit at end-of-life. However, considering the large area exposed to the space environment and the consequent high number of debris impacts per unit time, a high tether survivability to orbital debris is of primary importance. This paper provides an estimation of the number of fatal impacts per unit time and per unit length on a tape tether, using for the first time an experimental ballistic limit equation that was derived for tapes and accounts for the effects of both the impact velocity and impact angle. It has recently been shown that, tape tethers, as opposite to round wires, are more resistant to space debris impacts. It is shown that considering a tape tether with cross section 45 mm × 50 μm, the number of critical events due to impact with non-trackable debris is always less than 0.01/yr/km, being maximum for orbit inclination of $i = 90°$.

1. Introduction

The use of electrodynamic tethers (EDT) to de-orbit satellites from Low Earth Orbit (LEO) has been studied for more than a decade [1–8]. This technique could be particularly useful for defunct satellite removal as well as post mission disposal of end-of-life spacecraft from crowded orbital regions, thus reducing the chances of *Kessler Cascading* [9].

The core functionality of EDT systems depends on their survivability to Meteoroids and Orbital Debris (M/OD), and a tether can become itself a kind of debris for other operating satellites in case of cut-off due to particle impact. As regards tether survivability, space mission evidences are still conflicting after having a number of tethers deployed in space: in the SEDS-2 experiment (Small Expendable Deployer System) [10], a braided Spectra tether of 0.78 mm diameter and ∼19.7 km length was severed just after 3.7 days in orbit at 350 km altitude and the remaining 7.2 km length at the Delta end appeared to remain intact for the remaining 54 observable days of its orbit life until re-entry on 7 May 1994 [11]; this means there was 1 cut in 460 observable km-days of exposure. Unlike SEDS-2, a 2 mm diameter and 4 km long tether in circular orbit, at an altitude of 1022 km and inclination of 63.4°, survived about 10 years in the TiPS mission (Tether Physics and Survivability Experiment) [12]. It comes out the need of further investigating the resistance of tethers to M/OD impact, and in this framework several investigations have been carried out with reference to round-wire tethers. Such studies show that, depending on single strand or woven configuration of the wire, a particle with size from 1/6 [13] to 1/3 or 1/2 of the tether diameter [14] is enough to cut the rope. Other authors suggest to use 0.25 and 0.33 for this critical value [15,16], while [10] assumed that a tether fails if any part of an adequately large debris passes within 35% of the tether diameter. All these results are based on the assumption that any impact on the wire excavates a crater whose volume depends on the M/OD kinetic energy, and they indeed provide a first estimation of the ballistic limit of a tether. However, they model the impact damage as a crater and hence their application is not rigorously valid for tethers whose cross section is thin, e.g. tapes. Furthermore, the damage dependence from the impact velocity and impact angle is not considered (this latter parameter could be particularly important for tethers with non-axis-symmetric cross section, e.g. tapes).

Recently, tape tethers have been proposed for end-of-life satellite de-orbiting [4,17], and prototypes are currently under development in the framework of the European Commission FP7 BETs project (Bare Electrodynamic Tethers) [18,19]. In this context, to evaluate the survivability of tapes against M/OD impacts and compare it with that of round wires has become a necessity, and a comprehensive analysis on the subject has been initiated by Khan

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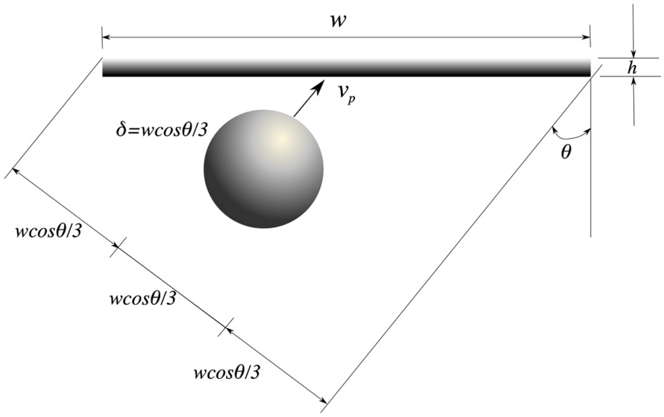


Fig. 1. Tape geometry and orthogonal projection along debris velocity direction.

and Sanmartin [20,21] extending to tapes previous results on the impact survivability of round wires. Assuming that a thin tape is severed by particles exceeding 1/3 of the orthogonal projection of the tape's section along the debris velocity vector (see Fig. 1) pointed out that tapes are less vulnerable to debris impact damage than round wires: for instance, at altitude 800 km and inclination 28.5°, tape tethers survivability was estimated to be about one and a half orders of magnitude higher than a round wire with same mass and length [20] because tape's cross section is reduced as impact obliquity is increased. Taking into account that fatal impact for tapes requires comparatively large and less abundant debris and tapes de-orbit faster, the overall survival probability of a tape during a de-orbit mission could be remarkably higher than an equivalent round wire.

In this framework, this paper provides a further step to enhance the accuracy of the risk predictions for tape tethers, thanks to the use of an experimental Ballistic Limit Equation (BLE) that was developed to account for effects not considered by the 1/3 geometric failure threshold i.e. the damage dependence from impact velocity and impact angle [22]. Such equation predicts the minimum particle diameter δ_m which produces a critical damage (cut-off) at a given speed, v_p, and impact azimuth angle θ, measured in the tape reference frame.

A few different debris-flux models are available based on the data gathered from the measurements over many years. The most prominent models of debris-flux are NASA's ORDEM2000 [23] and ESA's MASTER2009 [24]. Flux values predicted by ORDEM2000 are up to one order of magnitude higher than MASTER2009 in the significant diameter range of a few millimetres [25]. To be conservative, in our analysis we used ORDEM2000 flux data.

The remainder of this paper is organized as follows: Section 2 discusses the approach and the equations used for risk assessment on tapes; Section 3 reports the main calculation results: first, the new BLE used for tapes is compared to the classic, only-geometric, 1/3 failure criterion, then the fatal impact rate (number of severing impacts per unit time and per unit length) for a given thin aluminium tape (45 mm × 50 μm cross section) is presented for a range of LEO altitudes (250–1500 km) and two orbit inclinations (63° and 90°), based on the cumulative debris flux provided by the NASA's Orbital Debris Engineering Model (ORDEM2000) [23]. Conclusions are finally given in Section 4.

2. Debris impact risk on tape tethers

Debris flux is mostly considered to be horizontal on orbital plane. ORDEM2000 assumes orbital debris objects in circular orbits and only considers the horizontal component of the debris velocity. This is justified as the horizontal velocity component is about 6 to 11 km/s while the radial velocity component is generally less than 0.1 km/s [23]. The peculiar shape of tapes (huge length and comparatively smaller width/thickness) and tape being assumed vertical (gravity-gradient) as usual, simplifies the issue of tape moving in apparently horizontal debris environment. In our study, we only consider the orbital debris, avoiding the micrometeoroids which comes from every direction except the earth and thus might needs further consideration of elevation and calculation of surface flux. High survival probability requires low fatal-impact count N_c in a Poisson probability distribution

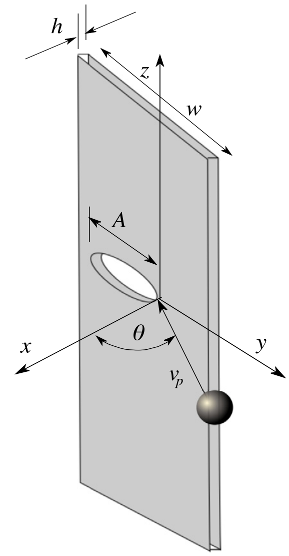


Fig. 2. Impact on tapes: geometry and reference frame.

$$P = e^{-N_c} \approx 1 - N_c \tag{1}$$

where $N_c << 1$. For a tether of length L deployed for a time period of Δt in orbit, one can roughly express the fatal impact rate per unit length, over some debris diameter $\delta > \delta_m$ range written as [20]

$$\frac{N_c}{L\Delta t} \cong \dot{n_c} = \int_0^{\frac{\pi}{2}} \frac{\mathrm{d}\theta}{\pi/2} I(\theta, v_p), \qquad I(\theta, v_p) = \int_{\delta_m}^{\delta_\infty} \frac{-\mathrm{d}F}{\mathrm{d}\delta}\, \mathrm{d}\delta\, D_{eff} \tag{2}$$

where $F(\delta, v_p)$ is the debris-flux and θ is the impact angle relative to the normal to the tape wide side. The upper bound δ_∞ in the δ-integral can be any large debris size. In our case, we selected $\delta_\infty = 10$ cm, that is the minimum limit of trackable debris for which conjunction analyses and avoidance maneuvers are possible. $D_{eff}(\delta, \theta)$ is the effective tape width calculated according to [27], see equation (3), and takes into account that debris which hit the tape off-center can sever the tether if they remove a critical amount of material A_{crit} from the tape: By definition, A_{crit} is the minimum quantity of material that must be removed to sever the tether, and $\delta_m(\theta, v_p)$ is just enough to produce a damage having size equal to A_{crit}.

$$D_{eff}(\delta, \theta, v_p) = w\cos\theta + \delta - \delta_m(\theta, v_p) \tag{3}$$

where w is the tape width (i.e. the front area per unit length) and $\delta_m(\theta, v_p)$ is the tape ballistic limit, i.e. the minimum debris size that is able to produce a tether critical damage (cut-off) at given impact conditions (angle θ and velocity v_p), measured in the tape reference frame, see Fig. 2 (left). Equation (3) states that a debris having diameter equal to δ and impacting the tether off-center can cause a critical damage provided that its overlapping with the tape is greater or equal to the critical value δ_m.

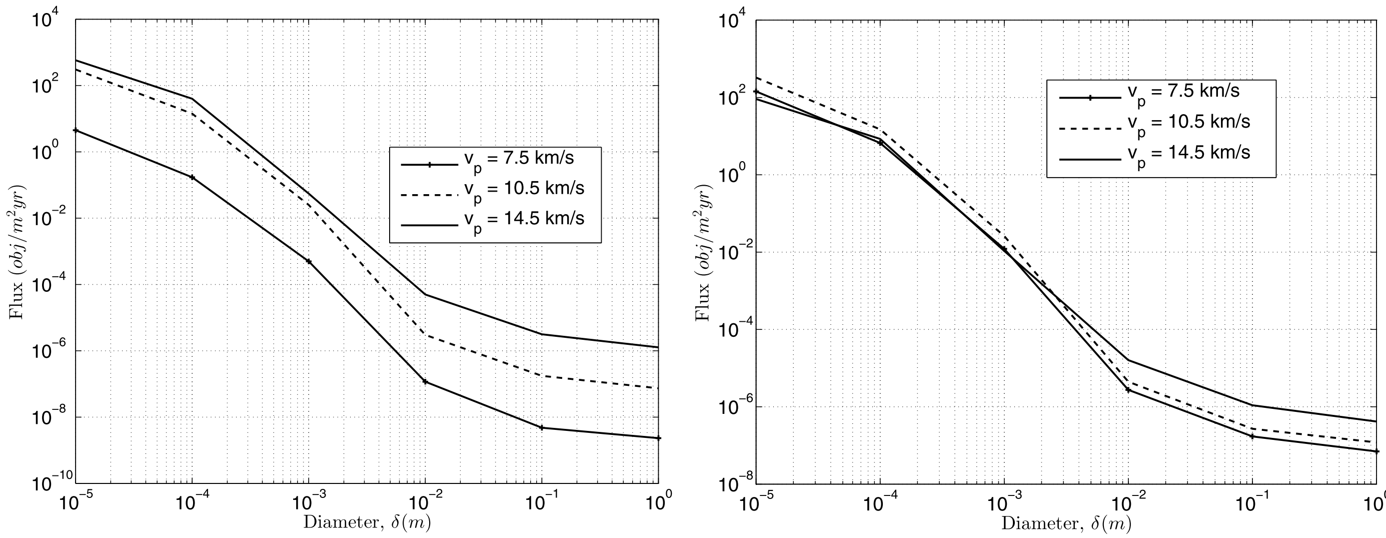


Fig. 3. Debris flux in function of particle size and velocity at $i = 90°$ (left) and $i = 63°$ (right) for a circular LEO orbit at 800 km altitude, using ORDEM2000 for Year 2013.

Note that A_{crit} defines the failure threshold, and could be related e.g. to the need of having a residual cross section capable of withstanding the maximum tensile load on the system, or the predicted electric current without overheating. Reducing substantially its cross-section area, the tape might survive a cut, but over weeks/months following the cut, carrying required current as planed in the mission design over such extremely reduced cross-section, would completely jeopardize the mission. Mission survival to a cut should thus require keeping, say, $A_{crit} \sim 0.8w$ of its cross-section, as assumed here.

Solving the integral in eq. (2) requires a suitable description of δ_m, i.e. a Ballistic Limit Equation (BLE). An empirical BLE for tape tethers has been recently developed [22], including the dependence of δ_m from the impact angle θ and impact velocity v_p:

$$\delta_m(\theta, v_p) = \frac{A_{crit}}{0.45}\left(\frac{v_p}{cos\theta}\right)^{-0.65} \tag{4}$$

where v_p is given in km/s. The above equation has been derived from experiments with impact angle not exceeding 80°, and is therefore sensible up to this limit; at θ close to 90°, the ballistic limit δ_m in eq. (4) tends to zero, which means that a debris of zero-diameter is able to cut the tether. The reason behind this inconsistency is that eq. (4) does not account for the progressive reduction of the debris damaging potential at impact obliquity approaching 90°: at such high impact obliquity the velocity component normal to the tape becomes so small that the shock waves originated by the impact have enough time to fragment the debris before it fully interacts with the tape. To account for this, the BLE was separated into two parts to prevent δ_m from decreasing below some fixed value δ_m^* at very high impact angles ($\theta \geq \theta_*$):

$$\delta_m(\theta, v_p) = \frac{A_{crit}}{0.45}\left(\frac{v_p}{cos\theta}\right)^{-0.65} \qquad \text{for } \theta < \theta_* \tag{5a}$$

$$\delta_m(\theta, v_p) \equiv \delta_m^* = \frac{A_{crit}}{0.45}\left(\frac{v_p}{cos\theta_*}\right)^{-0.65} \qquad \text{for } \theta \geq \theta_* \tag{5b}$$

In order to evaluate this piece-wise equation for different choices of θ_*, two values were selected for this threshold, i.e. $\theta_* = 80°$ (that is the experimental limit of eq. (4)) and $\theta_* = 85°$. This second value has been selected based on the work by [26], which shows that damage due to highly oblique impacts on thin aluminium targets (such an aluminium tape tether) does not increase anymore above $\theta \sim 85°$ if the impact velocity is ∼20 km/s. This velocity has been conservatively chosen because the largest part of the debris flux is below this threshold. Furthermore, at lower velocity θ_* is $< 85°$ and hence the experimental equation is within its range of validity. It may be noticed that [26] refers to semi-infinite plates, but this is approximately true also in our case (the target width is 45 mm and δ_m^* is around 2.3 mm at 20 km/s, i.e. around 5% of the width).

In summary, eq. (2) can be written as,

$$\frac{N_c}{L\Delta t} \cong \dot{n}_c = \int_0^{\theta_*} \frac{d\theta}{\pi/2} I(\theta, v_p) + \int_{\theta_*}^{\frac{\pi}{2}} \frac{d\theta}{\pi/2} I(\theta, v_p) \tag{6}$$

where,

$$I(\theta) = \begin{cases} I_1(\theta, v_p) = w\, cos\theta\, F(\delta_m(\theta)) + \int_{\delta_m(\theta)}^{\delta_\infty} F(\delta)\mathrm{d}\delta, & \text{for } \theta \leq \theta_* \\ I_2(\theta, v_p) = w\, cos\theta\, F(\delta_m^*) + \int_{\delta_m^*}^{\delta_\infty} F(\delta)\mathrm{d}\delta, & \text{for } \theta > \theta_* \end{cases} \tag{7}$$

3. Results

Eqs. (6) and (7) can be used to calculate the fatal impact rate of a tape tether, provided that the debris flux $F(\delta, v_p)$ and the ballistic limit $\delta_m(\theta, v_p)$ are known. In this paper, results are reported with reference to a 45 mm × 50 µm aluminium tape in circular orbits with altitude from 250 to 1500 km and two inclinations, i.e. 63° and 90°. The analysis is limited to high and moderately high inclined orbits because satellites are more numerous in such orbits, thus indicating that electrodynamic de-orbiting could be applied more extensively there. Furthermore, in highly inclined orbits the debris flux is higher and electrodynamic tethers are less efficient due to the magnetic field configuration: for these two reasons, considering highly inclined orbits is indeed more conservative than analyzing equatorial/near-equatorial orbits.

Considering the debris population, apart from orbit inclination and altitude, the flux $F(\delta, v_p)$ is a function of particle size and relative impact velocity. NASA's ORDEM2000 provides particle flux in six size bins ranging from 10 µm to 1 m, and in 23 velocity bins ranging from 0.5 to 22.5 km/s. Fig. 3 shows the debris flux on a circular orbit at 800 km altitude where the debris population is nearly maximum, as a function of both the particle diameter and speed, for $i = 90°$ (left) and $i = 63°$ (right).

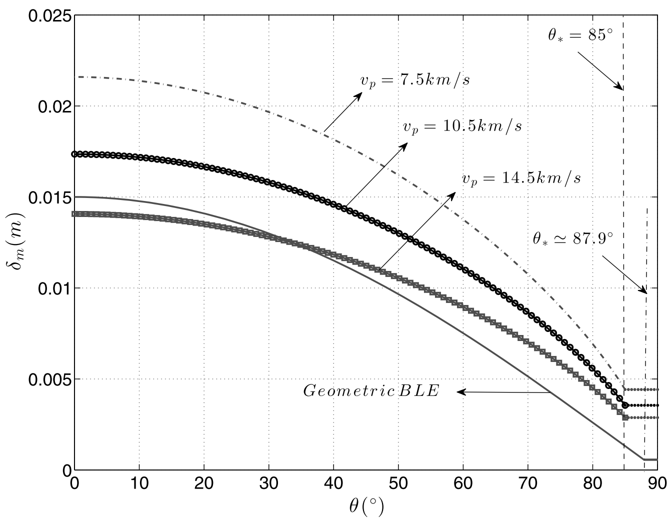


Fig. 4. Ballistic limit predicted by the new BLE (eq. (5a), eq. (5b)) and the Geometric BLE for a tape with width $w = 45$ mm and thickness $h = 50$ µm.

For both inclinations, it appears that the flux is maximum for particles with diameter in the range 10^{-5}–10^{-4} m and, in the case of velocity, flux is higher for $v_p \sim 14.5$ km/s at high inclined orbit, as can be readily verified from Kessler's debris velocity distribution formula [27]. On the other hand, for comparatively low inclined obit ($i = 63^\circ$) this is only true for larger debris, i.e. for diameter range $\sim 10^{-3}$–1 m. Another notable feature one can derive from Fig. 3 is that for highly inclined orbits all three curves are sparsely distributed, while in the case of the orbit with 63° inclination all the curves are densely situated, i.e. the variation of flux with debris impact velocity is higher for orbits which are highly inclined. These considerations point out the need of including the velocity dependence on flux calculations.

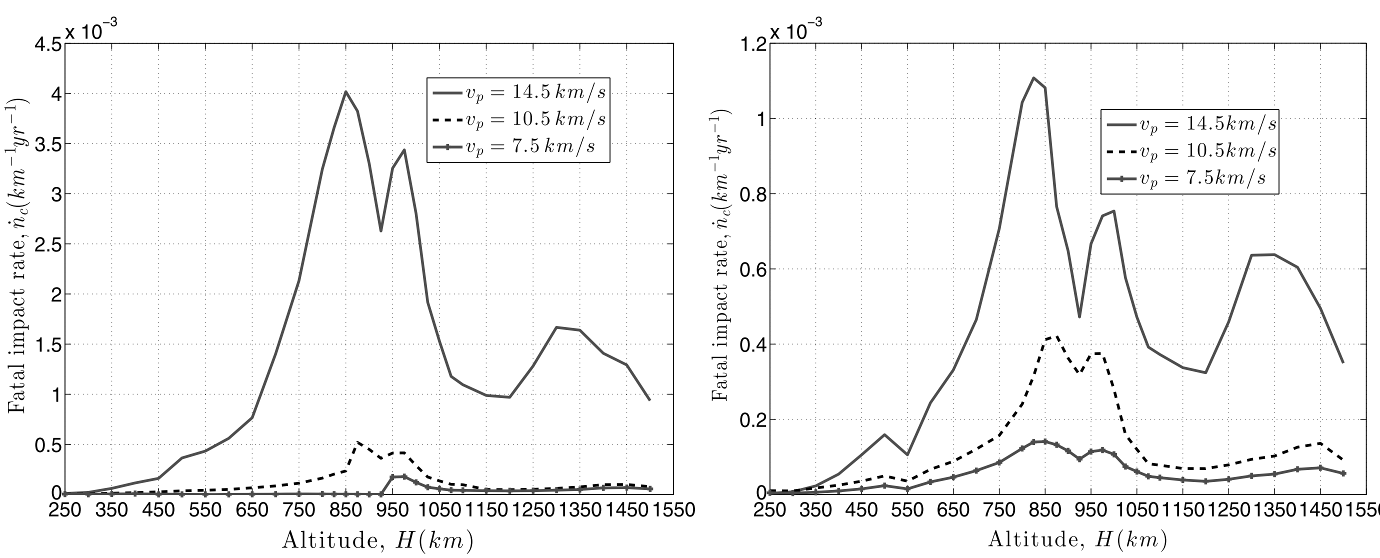
As regards the damage, the ballistic limit has been estimated with eq. (5a) and eq. (5b), that were derived from impact experiments on tapes, and incorporate the effects of both the impact angle θ and the impact velocity v_p. Fig. 4 presents a comparison between the new BLE (for three different values of v_p and $\theta_* = 85^\circ$) and the previous one [21], referred to as "Geometric" BLE in the figure as stated in section 1, "Geometric" BLE is based on the assumption that, independently from the impact velocity, a tape can be severed by particles exceeding 1/3 of the orthogonal projection of the tape's section along the debris velocity vector. Furthermore, unlike choosing a value for θ_* during glancing incidence, the "Geometric BLE" calculates it as, $\theta_{gl} \approx cos^{-1}\sqrt{4h/\pi w}$ [21].

Looking at Fig. 4, it appears that the new BLE is comparable to the geometric BLE for impact velocity ~ 14.5 km/s, and predicts debris critical values larger than those calculated with the geometric BLE for highly oblique impacts. This means that the geometric BLE overpredicts the tether failure rate, especially for non-orthogonal impacts, which are predominant; its employment for risk assessment therefore provides conservative results.

Finally, coming to tape tethers survivability, Figs. 5 and 6 show the fatal impact rate of a tape with width $w = 45$ mm and thickness $h = 50$ µm for two different θ_* values (80° and 85°). For both cases, the calculation has been done for two inclinations ($i = 63^\circ$ and $i = 90^\circ$) and 32 different altitudes from 250–1500 km. One might notice that there is a significant difference in the fatal impact rate depending on impact velocities for the case of high inclined orbit ($i = 90^\circ$) and this can rise as high as about three orders of magnitudes if velocity varies from 7.5 to 14.5 km/s. However, in the case of ($i = 63^\circ$) this difference can reach at most one order of magnitude.

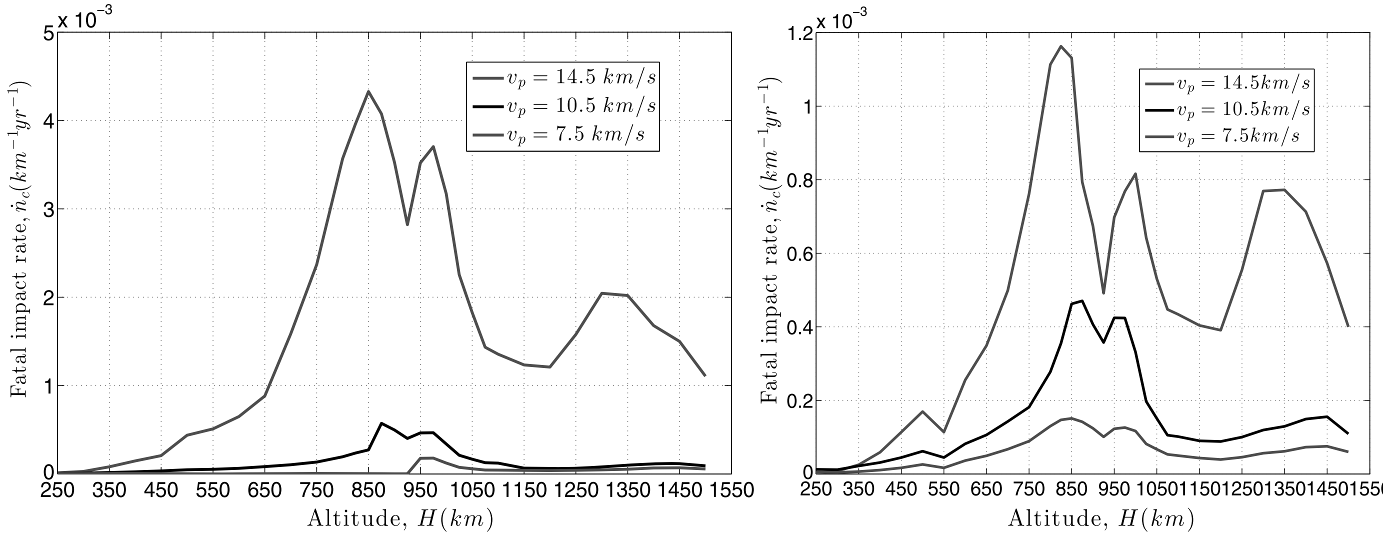
Comparing Fig. 5, it could be also noted that the fatal impact rate slightly increases from $\theta_* = 80^\circ$ to $\theta_* = 85^\circ$. This is because more debris with smaller size are incorporated in the failure flux with the highest value of θ_*. However, the increment is insignificant, showing that the sensitivity of results to θ_* is poor, thus avoiding the need of performing impact tests very close to 90°. Furthermore, this confirms that the failure rate of a tape is dominated by large debris.

Finally, Fig. 7 shows a summary plot of the total fatal impact rate considering all the possible impact velocities as a function of altitude, H. This figure indicates for each impact altitude, what is the expected cumulative failure rate for all debris diameters and all impact velocities in consideration. It appears that the total fatal impact rate is higher at high inclination ($i = 90^\circ$), and in both cases it is maximum between 850 and 1000 km. Considering a 10 km-long tape, it comes out that the number of critical events due to non-trackable debris is always less than 0.09/year at $i = 90^\circ$ and 0.06/year at $i = 63^\circ$. This is clearly a good result, since the numbers here denote the fatal impact rate at each particular



(a) $i = 90^\circ$ (b) $i = 63^\circ$

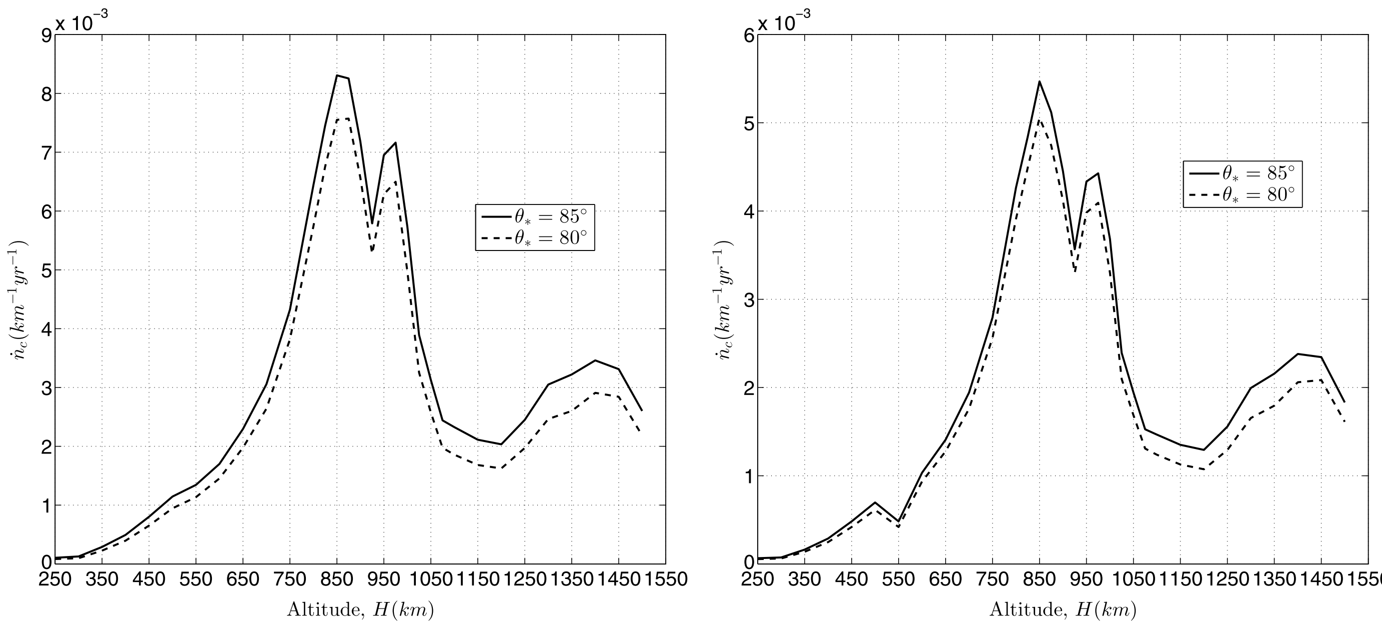
Fig. 5. (For $\theta_* = 80^\circ$.) Fatal impact rate of a tape as function of relative impact velocity and altitude, deployed at 63° and 90° inclined orbit, for Year 2013.



(a) $i = 90°$

(b) $i = 63°$

Fig. 6. (For $\theta_* = 85°$.) Fatal impact rate of a tape as function of relative impact velocity and altitude, deployed at 63° and 90° inclined orbit, for Year 2013.



(a) $i = 90°$

(b) $i = 63°$

Fig. 7. Fatal impact rate of a tape integrated on impact velocity as function of altitude, deployed at 63° and 90° inclined orbit, for Year 2013.

altitude. The duration of de-orbit missions may vary up to several months, depending on the satellite mass and its initial orbit altitude and inclination, hence making the overall survivability of the tape during the mission significantly optimistic.

4. Conclusions

This work presented an assessment of the debris impact risk to electrodynamic tape tethers for LEO spacecraft end-of-life deorbiting. The orbital debris flux for different impact velocities for a range of 32 LEO altitudes at two different inclinations, $i = 63°$ and $i = 90°$, was considered and the number of fatal impacts per unit time was calculated at each orbit altitude. To do this, a new experimental ballistic limit equation (BLE) for tapes was employed for the first time, taking into account the influence of the impact velocity and angle on failure. This new BLE provides more accurate predictions than classic geometric BLE for round wires, which are shown to be more vulnerable to debris impacts than tapes. With reference to a tape given cross section (width, $w = 45$ mm, thickness, $h = 50$ μm), the fatal impacts rate is estimated to be always below 0.009/year/km at $i = 90°$, and always below 0.006/year/km at $i = 63°$.

Conflict of interest statement

No conflict of interest.

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